FINAL CAPSTONE PROJECT

Computer Science and System Engineering

Sales Analysis Dashboard

using Superstore Dataset

Microsoft Power BI · DAX · Power Query · Kaggle Dataset

STUDENT	ROLL NO	PROGRAM	TOOL
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PROBLEM STATEMENT

The Business Challenge

1. Retail businesses generate thousands of daily transactions — making manual analysis nearly impossible.
2. Decision-makers lack real-time visibility into which products, regions, and categories are profitable.
3. Seasonal trends, discount impacts, and sub-category losses remain hidden in raw spreadsheet data.
4. Without an analytical tool, inventory and marketing resources are misallocated across regions.

Our Solution

Build an interactive Power BI Sales Analysis Dashboard that transforms raw Superstore transactional data into clear, actionable business intelligence — enabling real-time decision making for any stakeholder.

- ✓ Track KPIs: Sales, Profit, Orders & Margin
- ✓ Spot regional & seasonal performance patterns
- ✓ Identify loss-making sub-categories instantly
- ✓ Enable drill-down with interactive slicers

PROJECT OBJECTIVES

01 Data Import & Cleaning

Load the Superstore CSV into Power BI and clean it using Power Query (fix data types, remove duplicates, handle nulls)

02 Column Engineering

Create calculated Year, Month, and Quarter columns from Order Date for time-based analysis

03 DAX Measure Creation

Write 8 custom DAX measures: Total Sales, Profit, Orders, Profit Ratio %, Avg Order Value, and YoY Growth

04 Interactive Dashboard

Build a professional 1-page dashboard with 4 KPI cards and 5 interactive chart visualizations

05 Slicer-Based Filtering

Enable dynamic filtering using Region, Category, and Date Range slicers that update all visuals simultaneously

06 Business Insights

Derive and present 7+ meaningful business insights on sales trends, profitability, and regional performance

DATASET DESCRIPTION

~9,994	21	2014–2017	3
Total Rows	Columns	Date Range	Categories

Key Columns

Order Date	Time-based trend analysis
Sales / Profit	Primary KPI metrics
Category / Sub-Cat	Product segmentation
Region	Geographic performance
Customer Segment	Consumer vs Corporate
Discount	Impact on profit analysis

3 Product Categories

- Furniture
- Office Supplies
- Technology

4 Sales Regions

East · West · Central · South

Source: Kaggle — Sample Superstore Dataset (Free Download)

TOOLS & TECHNOLOGIES

Everything used to build this project — from raw data to polished dashboard

PB

Power BI Desktop

Main dashboard tool (FREE from Microsoft). Used to import data, build visuals, and publish reports. Industry-standard BI platform.

PQ

Power Query Editor

Built-in ETL tool inside Power BI. Used for all data cleaning steps: fix types, remove duplicates, create Year/Month columns.

DA

DAX Language

Data Analysis Expressions. Formula language to write custom measures: Total Sales, Profit Ratio, Sales Growth %, and more.

CS

Kaggle / CSV

Source of the Sample Superstore dataset — a real-world retail dataset with 9,994 rows and 21 columns of order data.

GH

GitHub

Version control and portfolio hosting. The entire project (PBIX, PPT, README, dataset) is uploaded as a public repository.

DASHBOARD OVERVIEW

Sales Analysis Dashboard — Power BI

$2.3M	$286K	5,009	12.5%
Total Sales	Profit	Orders	Profit %

Sales Over Time (Line Chart)

Sales by Category (Pie)

[INSERT YOUR POWER BI SCREENSHOT HERE]

Sales by Region

Profit by Sub-Category

Dashboard Layout

ZONE 1 — TOP ROW

4 KPI Cards: Sales, Profit, Orders, Profit %

ZONE 2 — MIDDLE

Line Chart + Pie Chart (sales trends & category share)

ZONE 3 — BOTTOM

Region Bar + Sub-Cat Profit + Top 10 Products

SLICERS — SIDE/TOP

Region | Category | Date Range interactive filters

Single page report · 1280×720px canvas · Dark Navy theme

DASHBOARD FEATURES

K

4 KPI Cards

Total Sales · Total Profit · Total Orders · Profit Ratio % — always visible at the top

L

Line Chart

Monthly sales trend over 4 years — spot seasonal spikes and YoY growth instantly

B

Regional Bar Chart

Horizontal bar chart comparing sales across East, West, Central, South regions

P

Category Pie Chart

Donut/Pie showing share of Furniture, Office Supplies, and Technology

S

Profit Sub-Cat Chart

Bar chart with red/green conditional formatting to identify loss-making sub-categories

T

Top 10 Products

Top N filter shows only the 10 highest-revenue products — driven by actual data

F

3 Interactive Slicers

Region · Category · Date Range — click to filter ALL charts simultaneously

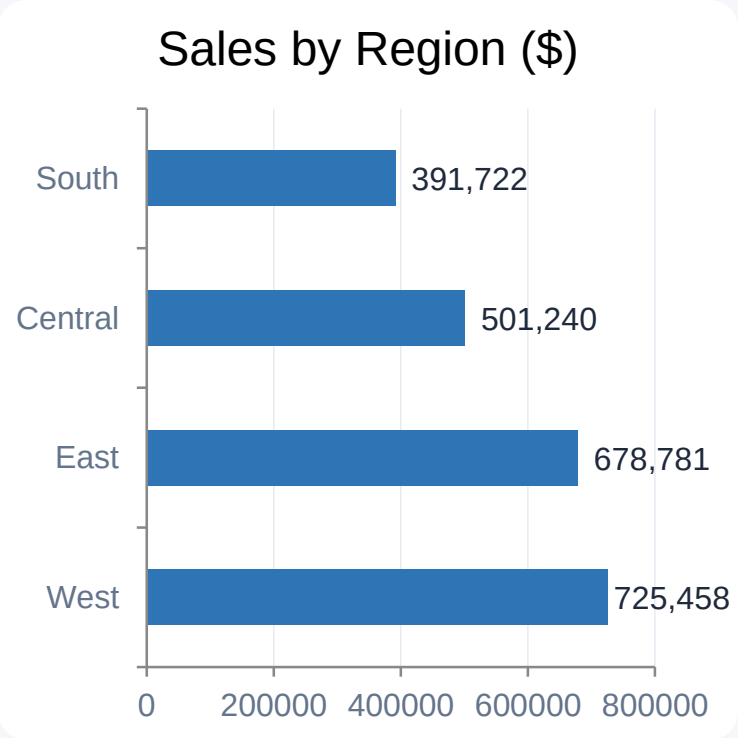
D

8 DAX Measures

Total Sales, Profit, Orders, Profit Ratio, AOV, Qty, Growth % — fully dynamic

KEY BUSINESS INSIGHTS

1. West Region leads in total revenue — contributing ~32% of overall sales
2. Technology category has the highest profit margin across all product lines
3. Tables & Bookcases record negative profit — caused by excessive discounting
4. November–December consistently shows the highest monthly sales (holiday season)
5. Discounts above 30% almost always result in a net loss on those orders
6. Consumer segment accounts for 50%+ of orders; Corporate has higher avg order value
7. Consistent 20–25% year-over-year sales growth observed from 2014 to 2017



WHAT MAKES THIS PROJECT UNIQUE

Technical decisions that set this dashboard apart from a basic report

U1

Dynamic Cross-Filtering

All 5 charts update simultaneously when any slicer is clicked — true end-to-end interactivity

U2

Error-Safe DAX with DIVIDE()

Every ratio measure uses DIVIDE(numerator, denominator, 0) to prevent divide-by-zero errors

U3

Time Intelligence (YoY)

SAMEPERIODLASTYEAR() function calculates year-over-year sales growth dynamically

U4

Conditional Profit Formatting

Sub-Category profit bars automatically turn red (loss) or green (profit) using Power BI rules

U5

Top N Filter on Products

Visual-level Top 10 filter shows only the highest-revenue products — fully data-driven

U6

Complete End-to-End Project

Full pipeline: Raw CSV → Power Query → DAX → Dashboard → Insights → Documentation → GitHub

FUTURE SCOPE & IMPROVEMENTS

F1

US Map Visual

Add a filled map showing state-wise sales using the built-in Power BI ArcGIS / Bing map visual

F2

Live Data Connection

Replace CSV with SQL Server or SharePoint for real-time, auto-refreshing dashboards

F3

Customer Drill-Through

Create a second page with customer-level analysis accessible via right-click drill-through

F4

Python/R Predictive Model

Integrate a sales forecasting model inside Power BI using the Python or R script visual

F5

Row-Level Security (RLS)

Implement RLS so each regional manager sees only their own region's data after login

F6

Power BI Service (Cloud)

Publish the dashboard to Power BI Service for scheduled refresh and stakeholder sharing via URL

Thank You · Samir Singh · Roll No: 2328117 · Computer Science and System Engineering